

The Arithmetic of Patience

How the source of finance decides what a UK sovereign endowment can be

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*This paper is the evidential companion to *The Long Asset*, which sets out the constitutional architecture and openly rests on the findings here. It is written to falsify its own working hypothesis as hard as to confirm it. The verdict at section 7 governs the constitutional paper's central claims: if the balance sheet proposition fails here, the institution described there should not be created.*

The two papers are deliberately separable. A reader may accept the constitutional architecture and reject the modelling, or the reverse, without having to reject both.

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A note on the evidence

The historical results in this paper, at section 5A and Annex B, are computed from the Jordà-Schularick-Taylor Macrohistory Database, a published academic dataset covering eighteen advanced economies from 1870. Every figure is reproducible from that source and the method is set out at Annex B.

They have **not been independently corroborated**. A second compilation of the same quantities may differ, and a single source cannot reveal an error it contains throughout. The historical findings are therefore presented as *measured and uncorroborated*: a weaker claim than established, and a stronger one than estimated. Section 7A states what an independent source would have to show for each to fail.

The forward estimates at sections 1 to 5 are assumptions, declared as such, and are a different kind of provisional figure: they await measurement rather than corroboration.

Summary

This paper tests whether the United Kingdom could convert sustained fiscal capacity into a long-duration financial asset, and finds that the answer depends entirely on which of two objectives the fund is given.

Two objectives, and they are not stages of one project. An *income* fund distributes an annual sum against the pension bill. A *balance sheet* fund accumulates an asset that improves the state's net position. The distribution rule chooses between them.

The income objective fails at any plausible contribution path. At £5bn a year and a 4.2% real return, the fund distributes £1.6bn at year 50, about 0.31% of the projected pension bill. The bill grows at roughly 2.7% real and a distributing fund does not. Reaching a tenth of it would require contributions of 0.6% to 1.2% of GDP every year. This is a finding, not an omission: the objective most readers assume is the point of such a fund is the one the arithmetic rejects.

The balance sheet objective survives. On the flat rule adopted here the fund reaches 13 to 33 points of GDP by year 50, against a debt path the OBR describes as unsustainable. The relevant test is not whether the fund grows but whether it grows faster than the cost of the borrowing that created it, so every figure here is reported as a net position rather than as a fund value.

Under debt financing the two objectives are mutually exclusive; under surplus financing the trade-off disappears. That converts the funding question from a menu of receipts into the binding constraint on the economic proposition this paper tests.

The historical record supports the surviving objective with a wide margin. Run against every complete rolling window since 1871 rather than a single assumed path, no forty-year or fifty-year window fails at the proposed rate, and one thirty-year window does. The realised financing spread is 3.68 to 7.40 points in every monetary regime since 1871, against a forward estimate of 0.4 to 1.2; the paper adopts the forward figure and states both.

A fund can be doing well and still look as though it is failing. In the worst window the fund held £155bn against £115bn contributed while standing £23bn behind the accumulated cost of the borrowing that built it. The fund was ahead of contributions and behind its financing benchmark in the same year. These are different quantities that peak in different years, and confusing them is the most likely way for a sound institution to be abandoned.

Diversification into bonds weakens the design on this criterion. A sixty-forty allocation fails three of 111 forty-year windows where a pure equity allocation fails none. The hedge costs more than it saves here, which is a measured result rather than a preference.

One observed joint state defeats independent stress testing, and this is the paper's methodological finding. Four years in 150 combine a real equity fall beyond 5% with a real financing cost above 5%, and three of them are a single episode that produces the only thirty-year failure in the record. A parametric stress test varies each leg across its plausible range separately, so it explores the space of *independent* departures and never constructs the conjunction, however large the individual shocks are made. Knowing to construct it requires having already found it. Realised sequences must therefore be run alongside parametric variation, not instead of it.

The evidence is measured and uncorroborated, and the paper says what would overturn it. Every historical figure is computed from a single published academic compilation, reproducible by any reader from the same source and the method set out here, and not independently confirmed. Section 7A states, for each principal finding, the observation in a second long-run dataset that would refute it. Two findings do not depend on corroboration at all, because they follow from definitions rather than from the level of any series.

Historical survival is not a forecast probability. Every window in the record is a window that happened. The record can show that no forty-year window failed; it cannot convert that into a chance of failure for a fund started next year, and this paper does not.

What is deliberately not here. Whether an institution could be built that would survive the decades this arithmetic requires is a constitutional question, and it is the subject of the companion paper, *The Long Asset*. The findings below say the economic proposition deserves that examination. They do not say the institution should be created.

1 Assumptions, and why each was chosen

All figures are in constant (real) terms. Contributions are paid at year end. The model is deliberately generous to the proposal wherever a choice was available, so that a negative result cannot be attributed to pessimistic inputs.

The portfolio constraint

The fund does not hold UK government debt. The usual reason given is that it would be inflationary. That is not quite the mechanism. A fund inside the public sector boundary holding gilts consolidates out entirely: the government issues debt to buy its own debt, the position nets to zero in the accounts, and the exercise costs transaction fees and achieves nothing. The objection is that it is null, not that it is inflationary.

The inflation concern is real but applies more widely, to domestic assets generally rather than to gilts specifically. It is the reason Norway's fund invests exclusively outside Norway: a sovereign fund buying domestic assets at scale bids up domestic prices and defeats the purpose of deferring consumption. A UK fund should therefore be constrained to non-UK assets, which is a stronger and more defensible rule than excluding gilts alone.

That constraint has a consequence for returns. Excluding the domestic bond market pushes the fund toward an equity heavy global allocation, which raises expected return and raises volatility.

Real return

The 125 year Dimson, Marsh and Staunton evidence gives annualised real returns of 5.2% for world equities, 1.7% for bonds and 0.5% for bills.

Allocation	Gross real	Net of 30bp	Comparator
60/40	3.80%	3.50%	Balanced, but includes domestic bonds
70/30	4.15%	3.85%	Approximately Norway's allocation
80/20	4.50%	4.20%	Central case here
100/0	5.20%	4.90%	Upper bound, unrealistic risk for a public fund

The central case is 4.2% real. This is higher than the 3.5% used before the domestic constraint was applied, and materially below the 5% used in early conversational sketches.

A constant rate is not a constant contribution, and the distinction governs how every table below should be read. The rate is held fixed; the base it is applied to is not. Retained returns enlarge the capital on which the next year's return is earned, and contributions continue to arrive, so the annual return in cash terms rises throughout the projection while the assumed rate never moves. The long horizon results are not produced by an optimistic rate. They are produced by an unremarkable rate applied to a base that grows for fifty years. Removing that mechanism alone, holding contributions, financing and the distribution rule fixed and earning returns on contributed principal only, does not shrink the result; it reverses it. The net position then peaks at £0.9bn in year 12, turns negative in year 18, and reaches -£2,823bn at year 100 against +£1,937bn. Compounding on retained returns is not a contributor to the proposition. It is the proposition. The financing side compounds in the same way at the government's real borrowing cost, which is why section 4 tests the difference between the two series rather than either alone.

Contribution paths

£2bn, £5bn, £10bn and £20bn a year, held constant in real terms. For scale, £5bn is approximately 0.18% of current GDP and £20bn approximately 0.73%.

The model assumes contributions continue in real terms for the full projection period, in every year, without interruption. This is a heroic assumption given the failure modes surveyed in the constitutional paper, and it is made deliberately: if the fund fails even under uninterrupted contributions, it certainly fails under realistic ones.

The denominator

State pension expenditure is currently around £138bn, approximately 5% of GDP. The OBR's July 2026 Fiscal Risks and Sustainability report projects it rising to around 9% of GDP by 2075-76 under the triple lock baseline, or around 7% under earnings indexation. Real GDP growth is assumed at 1.5%.

This is the most important input and the one most often omitted from endowment arguments. **The target is not stationary.** Under the baseline, real pension expenditure grows at roughly 2.7% a year: about 1.5% from GDP growth and about 1.2% from the rising share. A fund must outgrow that merely to hold its ground.

A separate proposal of the author's, replacing the triple lock with a personal allowance pegged to the full new state pension and uprated by a collar approximating price indexation, would produce a lower path than either OBR scenario, because a collar that tracks prices grows the pension bill more slowly than an earnings-linked or triple-locked rule. **The point stands independently of that proposal.** Any

uprating reform producing a slower-growing pension bill shrinks the liability this fund is sized against, and therefore reduces the contribution required for a given degree of offset. Nothing in this paper depends on which reform, if any, is adopted.

Drawdown rules

Rule	Specification
Flat	Nil to year 20; 3% of realised real return in years 21 to 30; 5% of realised real return thereafter
Rising	As above to year 30, then rising linearly to 60% of realised real return by year 60

2 The income test: can the fund pay for pensions?

£5bn a year, flat rule, at three real returns.

Year	2.5%	4.2%	4.9%
30	£0.2bn / 0.05%	£0.4bn / 0.11%	£0.4bn / 0.15%
50	£0.6bn / 0.11%	£1.6bn / 0.31%	£2.2bn / 0.42%
100	£2.4bn / 0.12%	£12.6bn / 0.64%	£24.1bn / 1.22%

Under the rising schedule, which is the fix designed to make the fund distribute:

Year	4.2% real	4.9% real
40	£4.2bn / 1.04%	£5.7bn / 1.41%
60	£21.7bn / 3.18%	£31.3bn / 4.58%
75	£30.0bn / 2.95%	£44.4bn / 4.36%
100	£49.4bn / 2.49%	£76.8bn / 3.87%

Two findings.

The flat rule cannot deliver income. Drawing 5% of a 4.2% real return is drawing 0.21% of the fund. The fund compounds impressively and distributes almost nothing, indefinitely.

The rising schedule delivers income, then loses the race. The share of the pension bill peaks around year 60 at roughly 3% and declines thereafter, because a fund distributing 60% of its return grows more slowly than a pension bill growing at 2.7% real. Under the most favourable combination modelled, £10bn a year at 4.9% real, the peak is 9.6% around year 60, still declining afterwards.

To reach 10% of the projected pension bill by year 50 requires contributions of roughly 0.6% to 1.2% of GDP every year, depending on returns. The UK spent approximately 2.1% of GDP on defence in 2023-24.

The income objective fails. No plausible contribution path funds a meaningful share of the state pension.

3 The balance sheet test: can the fund offset borrowing?

This is a different question and it produces a different answer, because it compares a stock to a stock rather than a flow to a fast growing flow.

Fund value as a percentage of GDP, rising schedule:

Contribution	Return	Year 30	Year 50	Year 75
£5bn	4.2%	6.6%	11.6%	14.4%
£5bn	4.9%	7.5%	14.1%	18.3%
£10bn	4.2%	13.3%	23.3%	28.9%
£10bn	4.9%	14.9%	28.1%	36.6%

That table is computed on the rising schedule, which section 6 concludes should be dropped. On the flat rule the paper adopts, the same four cases give 13.3 to 32.8 points at year 50, and the divergence widens with horizon: the £5bn at 4.2% case reaches 14.4% of GDP at year 75 on the rising schedule and 26.8% on the flat rule, because the rising schedule's distributions bite hardest late, when they are taken from the largest capital base. Against PSNFL of approximately 84% of GDP today, and a debt path the OBR projects as unsustainable from the 2040s, an asset worth 13 to 33 points of GDP is materially significant. On the balance sheet test the proposition looks an order of magnitude better than on the income test.

But the gross figure is not the answer. Under PSNFL, borrowing to acquire financial assets is neutral by construction. The fund improves the balance sheet only to the extent its return exceeds the cost of the debt raised to fund it.

4 The net position test

Fund value minus the debt incurred to make the contributions, both compounding in real terms. Contributions of £5bn a year, real financing cost of 3.5%, which is a 30 year gilt around 5.5% nominal against 2% inflation.

Under the **rising** schedule:

Year	Fund	Debt	Net	Net as % GDP
30	£287bn	£258bn	£29bn	0.7%
50	£676bn	£655bn	£21bn	0.4%
75	£1,217bn	£1,743bn	minus £526bn	minus 6.2%
100	£2,000bn	£4,313bn	minus £2,313bn	minus 18.9%

Under the **flat** rule:

Year	Fund	Debt	Net	Net as % GDP	Payout
30	£287bn	£258bn	£29bn	0.7%	£0.3bn
50	£776bn	£655bn	£121bn	2.1%	£1.6bn
75	£2,272bn	£1,743bn	£529bn	6.3%	£4.6bn
100	£6,250bn	£4,313bn	£1,937bn	15.8%	£12.6bn

This is the central finding of the section.

A debt financed fund can compound into a genuine net asset, or it can distribute income, but it cannot do both. Distribution stops the fund compounding while the debt continues compounding, so the rising schedule turns the net position negative somewhere between years 50 and 75. The flat rule preserves the net position precisely because it distributes almost nothing.

The two purposes are not complementary. They are alternatives, and the drawdown rule is the instrument that chooses between them.

If contributions come from genuine surplus rather than borrowing, no debt accrues and the trade-off disappears: at £5bn a year and 4.2% real, on the rising schedule, the fund reaches 11.6% of GDP by year 50 while distributing £11bn a year, and every pound is a net asset.

5 The financing spread

At the time of writing, long dated gilt yields are in the region of 5% to 5.8% nominal, though aggregator sources disagree materially on the same date and current levels should be taken from the DMO directly. Against long run inflation of 2%, the real financing cost is roughly 3.0% to 3.8%.

The central portfolio return, net of costs, is 4.2%.

The expected real spread is therefore approximately 0.4 to 1.2 points. Positive, unlike the near zero spread implied by a 60/40 portfolio including domestic bonds, but thin. And it is entirely an equity risk premium harvested with borrowed public money, which is precisely the transaction the New Zealand critique identified: a fund that is effectively fully leveraged, where taxpayers may not be compensated for the risk they are assuming.

The realised record contradicts that estimate, and the discrepancy is itself a finding. Computed on annual data from 1871 to 2020, in real sterling, the spread between world equity returns and the realised UK long-term government rate is 4.36 points under the gold standard, 3.68 through the wars and interwar period, 7.40 under Bretton Woods and 5.44 since floating. The narrowest realised spread in a century and a half is three times the top of the range above, and it occurs in the era containing two world wars and the Depression.

Both figures cannot be right, and the paper does not resolve which. What the comparison establishes is that **the financing assumption, not the return assumption, is where the uncertainty sits**: sensitivity analysis on the same data indicates that the real financing cost moves the result roughly five times as much as switching the equity weighting basis. A forward estimate built from spot gilt yields and a realised estimate built from a century and a half of outturns differ by a factor of three to six, and the paper's declared dependency on the financing proxy at section 12 is therefore the dominant open question rather than one of several.

These historical figures are computed from the Jordà-Schularick-Taylor Macroeconomy Database and have not been independently corroborated; see the note on evidence.

A thin positive expected spread over a century is not nothing. Whether it survives a bad thirty year sequence is a question this paper originally could not answer, because the deterministic model contained no sequence of returns risk.

It can now be answered, and the answer is not the one anticipated. Taking every complete rolling window in the 1871-2020 record rather than a single start date: of 121 thirty-year windows, one fails at the proposed distribution rate, beginning in 1902 and examined below. Of 111 forty-year and 101 fifty-year windows, none fails.

5A The historical tests

Section 5 sets out a forward estimate of the spread and section 4 a deterministic net position. This section reports what the realised record does to both. Each test below is framed as a question whose answer could defeat the proposal, not as a demonstration that it works. Where a test was survived, the margin is given; where a boundary exists, it is located.

All figures are computed from the Jordà-Schularick-Taylor Macrohistory Database, 1871 to 2020, in real sterling. They have not been independently corroborated. The construction, the declared analytical choices and the reproduction method are at Annex B.

5A.1 Where does the proposition cease to work?

The relevant boundary is not a pass or fail. It is the maximum share of real return the fund can distribute while its terminal net position remains positive. Below that share the proposition holds; above it, the fund distributes faster than it compounds against the debt.

horizon	windows	worst	median	best
30 years	121	0.0%	51.3%	94.3%
40 years	111	13.2%	49.2%	81.3%
50 years	101	21.6%	50.5%	71.0%

The proposed flat rule distributes 5%. At forty years the worst historical window tolerates 13.2%, so the margin is roughly two and a half fold at the worst start and ten fold at the median. A rising schedule distributing 60% exceeds the boundary at every start in the record, which is the section 4 finding confirmed on realised data rather than on a central assumption.

What would defeat the proposal. A second compilation moving the worst-case boundary below 5% would defeat the design rather than narrow it. Nothing in this record approaches that: the lowest boundary at any horizon of forty years or more is 13.2%.

The exception, and it is instructive. One thirty-year window fails outright: 1902 to 1931, containing the First World War, the collapse of 1920-21 and the onset of the Depression, over which the realised spread was 1.3 points against a record median above four. It recovers at forty and fifty years. The failure is therefore not evidence that thirty years is the wrong horizon but that a thirty-year window can contain an era in which the spread substantially disappears.

5A.2 Would relaxing the portfolio constraint improve robustness?

Section 1 constrains the fund to equities on an argument about domestic gilts consolidating to nothing. That argument is correct and answers only the obvious case. The harder objection is diversification into international bonds, which is a genuine asset class with a positive long-run real return and which behaves well in precisely the episodes that threaten an equity fund.

Tested directly, across every complete forty-year window:

equity weight	real return	margin over financing	terminal failures
100%	6.92%	5.09pp	0 of 111
80%	6.22%	4.39pp	0 of 111
60%	5.45%	3.62pp	3 of 111

The hedge works. In 1929 and 1930 world bonds returned 6.5% and 9.6% in real terms while equities fell 7.0% and 14.6%. And it costs more than it saves: over the full record world bonds returned 2.67% real against a financing cost of 1.83%, a margin of 0.84 points where equities earned 5.09 over the same cost. Forty years of thinner compounding outweighs one decade of cushioning.

The three failures under a sixty-forty allocation share a property that is not their date. Each terminates in a year of exceptionally high real financing cost, and one of them terminates in 1931 itself. **The hedge worked, in the episode it exists for, and lost anyway.**

The defensible statement is therefore narrower than *equities are better*: the principal alternative was modelled directly, and on the criterion adopted here, over these horizons, diversification into bonds weakens rather than strengthens the design. A different criterion, minimising interim drawdown, say, rather than maximising terminal survival, would rank them the other way, and a fund with liabilities falling due in downturns would be right to choose differently.

5A.3 What happens inside the bad sequences?

Terminal survival conceals the path. Three quantities distinguish what the fund experiences from what the public accounts record, and they are not interchangeable.

Fund value cannot be negative. An unleveraged equity portfolio's worst outcome is total loss. Across every forty-year window the fund's own worst peak-to-trough drawdown is **50.7%**, with a median worst drawdown of **31.0%**. A sovereign equity endowment should expect to lose roughly a third of its value at some point in any forty-year period, and half in the worst case. That is the honest statement of market risk and it belongs in any public presentation of the proposal.

Net position can be negative, and frequently is. Of 121 thirty-year windows at the flat rule, 25 are continuously solvent, 95 dip below zero at some point, and 1 fails terminally. Dipping is the normal case.

The two peak in different years. Fund drawdown and net shortfall coincide in under 2% of windows. In the worst historical window the deepest fund drawdown falls in 1948 and the deepest net shortfall in 1931, because by 1948 the fund was large enough that a 31% fall still left it far ahead of its financing cost.

A single figure makes the distinction concrete. At the 1931 trough of the worst window, the fund held £155bn against £115bn contributed, while standing £23bn behind the accumulated cost of borrowing that sum. **The fund was £40bn ahead of contributions and £23bn behind its financing benchmark simultaneously.** Neither statement is misleading; they answer different questions, and conflating them produces the impression of a fund in deficit, which an unleveraged portfolio cannot be.

5A.4 Is the favourable financing behaviour real, or accidental?

Sections 4 and 5 treat the financing cost as an independent parameter, varied on its own. That treatment is unsafe, and the reason is narrower and more interesting than a general correlation.

There is no general adverse correlation. The relationship runs the other way. Across the 150-year record the correlation between the real world equity return and the real UK financing cost is **+0.291**, and within the 31 deflationary years, a deflationary year being one of negative UK consumer price inflation, it is **+0.239**. Both are positive. Deflationary years are on average *better* than average for real equity returns, a mean of +10.33% against +8.10% across all years, because falling prices raise the real return on nominally priced assets as well as raising the real cost of nominal debt.

An earlier draft of this section asserted that the two legs are negatively correlated in the state that matters. That claim is false on this record and is withdrawn.

What defeats the independence assumption is not a correlation but a single joint state. Four years in 150 combine a real equity fall beyond 5% with a real financing cost above 5%:

year	real equity	real financing	UK inflation
1929	-7.0%	+5.5%	-0.9%
1930	-14.6%	+7.5%	-2.8%
1931	-17.3%	+9.2%	-4.3%
1987	-5.4%	+5.9%	+3.2%

Three of the four are one episode, and that episode produces **the only thirty-year failure in the record**. It is also the terminal year of the deepest failure under the sixty-forty allocation at §5A.2.

That is sufficient, and the general claim was never needed. A parametric stress test varies each leg across its plausible range separately. It therefore explores the space of *independent* departures and never constructs the conjunction, however large the individual shocks are made. One observed joint state that is decisive for the result is enough to show the method can miss the path that matters. No claim about the frequency or the sign of the general relationship is required.

Two consequences follow, and the second is the one the paper's own method should adopt.

The pure-equity structure is exposed in this specific state. A fund holding bonds is partly hedged against a deflationary collapse; a fund holding only equities is not. That is the measured price of the allocation chosen at section 1, and §5A.2 shows the hedge costs more than it saves on this criterion, but the exposure is real and belongs on the record as a price rather than being omitted.

Realised sequences must be run alongside parametric variation, not instead of it. The sequence contains the joint case as a fact. The parametric method would have to know to construct it, and knowing to construct it requires having already found it.

5A.5 What must the institution be able to carry?

The historical stress translates into three distinct requirements, in three different dimensions. At a contribution of £5bn a year in real terms, taking the worst window in the record:

requirement	quantity	dimension	who owns it
standby facility	£23bn	a stock	Treasury balance-sheet capacity
exposure carried	£65bn-years	a stock × time	none
carrying cost	£0.64bn	a flow, integrated	a spending settlement

The middle row is an area under a curve rather than a sum of money and is not a facility requirement; it is the input to the third. The facility the government must actually arrange is £23bn, and the cost of carrying that exposure at the window's own real financing rate is under a billion pounds in total.

Two further figures matter for how long the arrangement must persist. The deepest single-year shortfall and the longest run of consecutive negative years fall in different windows, so no single path exhibits both. And in the worst window the net position does not become durably positive until **year 25**, in a fund that terminates strongly positive.

That last figure is the institutional problem stated as a number, and it is what the companion paper's constitutional architecture exists to address.

5A.6 When does net position become informative?

Every complete forty year window in the 1871 to 2020 record was run at the proposed distribution rule, and the fund's net position recorded at each year of its life. All 111 windows succeed terminally. For each fund age the calculation reports four measures: the correlation between position at that age and the terminal outcome, the lowest position still compatible with eventual success, the share of successes still negative, and the share of positive readings that later return to negative.

The hypothesis the profile rejects. The sole thirty year failure begins in 1902, and its durable positive crossing occurs at year 25, which suggested that year 25 might mark the age at which position becomes informative. Profiling by fund age rejects that inference: no discontinuity occurs at year 25 on any measure. Informativeness develops gradually instead. Of paths showing a positive position at age 15, 26% later return to negative; at age 25, 11%; at age 30, 3%; by age 33, none. No ultimately successful path is negative after age 33, and the latest such path begins 1899 and terminates strongly positive.

What the profile can and cannot establish. Every window in the record succeeds terminally, so it answers how often a successful programme still looks unsuccessful at a given age. It cannot answer the converse, how often an unsuccessful appearance is real, because there are almost no failures against which to test a discriminator. The record can therefore establish the age at which position becomes more informative among successful paths. It cannot establish a threshold at which position constitutes evidence that the proposition has failed. What an institution may do with that asymmetry is a question for the constitutional paper, not for this one.

6 Reconsidering the proposed drawdown rule

The flat rule was assessed at 3.2 against the income objective and failed absolutely. Against the balance sheet objective it is the only rule that works.

That reverses the earlier judgement. The rule was not wrong. The objective it was being judged against was wrong. A fund whose purpose is to offset borrowing should distribute almost nothing, because every distribution is a pound that stops compounding against a debt that does not stop compounding.

The proposed rule is therefore kept and the stated objective changed, rather than the reverse.

7 Verdict

The working hypothesis stated that a UK endowment can become fiscally material only over several generations. That is not supported in the form stated, but the reason is more interesting than a timescale problem.

1. **The income objective fails at any plausible contribution path.** The pension bill grows at roughly 2.7% real. A distributing fund does not.
2. **The balance sheet objective succeeds, and succeeds materially.** A fund worth 13 to 33 points of GDP by year 50, on the flat rule this paper adopts, is significant against a debt path the OBR calls unsustainable.
3. **The two objectives are mutually exclusive under debt financing.** The drawdown rule chooses between them, and the choice cannot be deferred.
4. **Under surplus financing the trade-off disappears,** which converts the funding question from a menu of receipts into the binding constraint on the economic proposition this paper tests.

7A What would overturn these conclusions

Each principal finding is stated with the observation that would refute it. These are commitments made in advance of any independent test, and they are the substance of the invitation at Annex C: any institution holding a second long-run dataset can perform them.

finding	what would overturn it
The realised spread is 3.68 to 7.40 points in every monetary regime since 1871	A second compilation showing a long-run spread below roughly 2 points, which would restore the forward estimate at section 5 and with it the original judgement that the margin is thin
No complete forty- or fifty-year window fails at the proposed distribution rate	Any such failure in a second compilation. The honest revision would then be a probability rather than a universal, and the paper would report the failure rate rather than its absence
The worst-case boundary at forty years is 13.2% of real return	A second compilation moving that below 5%, which would defeat the proposed rule rather than narrow it
Diversification into bonds weakens the design on this criterion	A second compilation in which a sixty-forty allocation produces fewer terminal failures than a hundred per cent equity allocation
Independent parametric stressing of the two legs is unsafe, because the record contains a decisive joint adverse state	A record in which no joint state of falling equities and high real financing costs is decisive for any outcome. Note this is not overturned by the absence of a general correlation: none exists, and §5A.4 does not rely on one

Two findings are not contingent on corroboration, because they do not rest on the level of the series. That fund value cannot be negative follows from the portfolio holding no leverage. That fund drawdown and net shortfall are distinct quantities peaking in different years follows from their definitions. A second compilation could change the magnitudes and could not change either.

And one dependency dominates the rest. Sensitivity analysis on this data indicates the real financing cost moves the boundary roughly five times as much as switching the equity weighting basis. A market-observed gilt series matched to the assumed issuance would therefore do more to settle these conclusions than a second equity compilation would, and unlike a second equity compilation it is obtainable from open sources. The equity corroboration gates what may be claimed; the financing series gates what the answer is.

What this settles for the constitutional paper

- **The choice of objective is not a framing preference.** It is determined by this arithmetic. The fund can be a material public asset or a material source of pension income, and under debt financing those are alternatives rather than stages.
- **The flat rule survives and the rising schedule does not**, except under genuine surplus financing, where no debt compounds beneath the fund and the trade off dissolves.
- **The funding question is not a menu.** The source of finance determines which objectives are available. Under debt financing only the balance sheet objective survives, and only under a low distribution rule; genuine surplus makes both available. A receipt that would otherwise have reduced borrowing is debt financing whatever it is called.
- **The decisive problem is unchanged, and is now better founded.** Preserving the institution through a long invisible period remains the constitutional difficulty, and the invisible period is shown here to be a consequence of the distribution rule rather than a fact of nature.

Annex A: model and reproducibility

Contributions at year end, drawdown taken from realised real return before reinvestment, no tax, no rebalancing drag, no sequence of returns risk, contributions never interrupted.

Every one of those simplifications was assumed to flatter the proposal. One did not. Adding sequence of returns risk, by running the proposal against every complete rolling window in the historical record rather than a smooth central assumption, left it standing at every forty and fifty year horizon without exception. The instinct that the model's conveniences were self-serving was correct in general and wrong about the one that could be tested. The remaining simplifications are untested and should still be read as favourable.

The deterministic model establishes a conditional result: under the central return path, restrained drawdown eventually produces a positive net balance sheet position. It does not establish that the architecture is likely to do so across plausible sequences of returns. Those are different claims, the first arithmetic and the second risk analysis, and a terminal value model could show a positive outcome at year 75 while concealing fifty or sixty years in which accumulated debt exceeded the asset and the institution remained exposed to political abandonment throughout.

This paper originally specified a stochastic model with return sequencing to answer four questions. **Three of them have since been answered empirically, by the historical testing at section 5A,** and the position on the fourth is more interesting than an outstanding task:

output originally specified	status
the distribution of first crossing years	answered. Balance sheet crossing at a median of ten years, range two to thirty-four
cumulative years spent below zero	answered. Of 121 thirty year windows, 25 are continuously solvent, 95 dip below zero, one fails terminally
the probability of a sustained positive net position	answered in the form the evidence permits. Of positions positive at age 15, 26% later return to negative; at 25, 11%; at 30, 3%; by 33, none
the probability of ever crossing into a positive net position	cannot be estimated, by any method this evidence supports

The fourth is the one worth dwelling on. Every complete window in the record crosses. A sample containing no failures cannot yield a failure rate, and a stochastic model asked for one would supply a number drawn from the tails of an assumed distribution rather than from the data. That number would be more precise than the evidence and less informative than saying plainly that no historical window failed.

And a stochastic model of the obvious kind would make the analysis worse rather than better. Section 5A.4 establishes that the only historical failure arises from a joint state in which equity returns fall while real financing costs rise, and that no general correlation exists to reproduce it. A simulation drawing the two legs independently, which is the standard construction, cannot generate that state at all, and would therefore report a lower failure rate than the historical record contains. The requirement, if it is ever met, is for a model that resamples joint observations rather than parameterising the legs separately; and a model resampling joint observations from this record is doing something close to what the rolling windows already do.

So the outstanding item is narrower than originally stated. Not stochastic verification of the proposition, which the historical testing has now largely supplied, but a bootstrap preserving the joint

structure, which would extend the number of paths without supplying new information about the distribution they are drawn from. It is worth doing and it is not what the conclusions are waiting on.

The fourth output is named for what it measures. It is not an investment statistic; it is a measure of institutional credibility. Two simulations may both cross into positive territory in year 48 with similar expected values, one staying positive thereafter and one oscillating either side of zero for twenty years. Economically they are close. Institutionally they are different institutions. In the first, every Chancellor inherits a fund that can be defended as strengthening the national balance sheet. In the second, every downturn revives the argument that the project has failed and should be wound up. A long early period of weak returns matters disproportionately, since the debt compounds at a contractual rate while the asset base is still small and has no time to recover.

Annex B: historical back test

Purpose, stated narrowly. This annex does not estimate what a future UK endowment would earn. It asks whether the balance sheet proposition would have survived one actual half century of world equity returns, including its crashes, recoveries and prolonged weak periods. It establishes a realised path, not a probability. Probability belongs to annex A.

Method

Annual observations and annual cash flows. For each year from 1975 to 2025:

1. add the contribution for that year, constant in real terms or as a constant share of GDP;
2. apply the actual world equity total return for that year, with the timing convention stated;
3. reinvest all income;
4. accrue the debt financing liability at the **actual** real financing rate for that year;
5. record fund, debt and net position;
6. and at the end, report first crossing, any reversions below zero, cumulative years below zero, and the durability of the final crossing.

This makes the annex a historical analogue of annex A rather than a decorative illustration.

Four guards on the specification

1. Real returns with real terminal values, and real contributions. A nominal return must produce a nominal terminal value. If the annex reports in 2025 pounds it uses real returns throughout. Contributions must be constant in real terms or as a share of GDP: £5bn in 1975 is roughly £42bn today, so a flat £5bn nominal from 1975 would be a radically front loaded programme rather than the historical equivalent of the proposal.

2. World, not US. A US only series tests an unusually successful national market after the fact. The world series is the central case because it avoids selecting the eventual winner. The US result appears as an explicitly labelled upper historical comparator and cannot carry the conclusion.

3. Currency reported separately, not embedded. Present investment performance in a common currency, and unhedged sterling separately to show the contribution of currency movement. Sterling fell from around \$2.30 to around \$1.30 over the period, and no part of the paper may imply that past sterling depreciation is a repeatable source of return.

One caveat on the hedged alternative. A synthetic hedged sterling series for 1975 to 2025 is not a clean counterfactual, because hedging cost tracks interest rate differentials, and UK rates sat far above US rates

for long stretches of the period. The embedded carry would be substantial and would itself need explaining. A local currency or common currency world index is the cleaner basis for isolating investment performance; a constructed hedged sterling series is not.

4. Both legs need actual series, not just the asset leg. Step 4 above says actual, and holds to it. Real gilt yields were negative through the late 1970s, strongly positive through the 1980s and 1990s, and negative again for much of 2010 to 2021. Compounding the debt leg at a flat 3% real while running the asset leg on actual returns would repeat, on the liability side, exactly the error the annex exists to avoid on the asset side.

A fifth guard: the start date

Beginning in 1975 starts immediately after the 1973 to 1974 crash, which is close to the best available entry point in the period. Choosing it after the fact is the same class of error as choosing the US market after the fact.

The annex should therefore run **rolling start dates**, for example every start year from 1960 to 1975, and report the distribution of outcomes across them. This is cheap once the series is in hand and it converts a single anecdote into a small sample. If the balance sheet proposition survives every start date it is a materially stronger finding than if it survives one.

Reporting, in three layers

1. the actual 1975 to 2025 world equity path, as the central result;
2. a US comparator, labelled as an upper historical outcome rather than a central series;
3. the deterministic mean return result, shown only as a reconciliation demonstrating how much the actual sequence changed the outcome.

For that reconciliation, the deterministic figures are: constant real contributions of £5bn over 51 years at 6.5% real gives approximately £1.83tn, against debt compounding at 3% real reaching approximately £586bn, for a net position of roughly £1.25tn. The historical series result will differ. **That difference is not noise. It is the whole test.**

The test has now been run, and it moved in the proposal's favour on both legs. Over 1975-2020 the realised world equity return in real sterling was 8.02% against the 6.5% assumed, and the realised UK real financing cost 2.38% against the 3% assumed. The paper anticipated the test correctly and predicted the wrong direction.

Two qualifications belong with that, and the second is the more important. The realised sterling return embeds four decades of sterling depreciation, which annex B warns should not be treated as a repeatable source of return, so hedged and unhedged constructions must both be reported. And the 1975 start is not a flattering one: across the full 1871-2020 record the modelled window sits below the historical median on every horizon, which removes the suspicion raised at section 12 that the chosen period favoured the proposal.

What a positive result would and would not establish

If the actual world sequence produces a durable positive net position despite the severe early and intermediate drawdowns, the annex supplies one historically realised path under which the debt financed, low drawdown institution passed its own test. That is more than illustration. It would materially strengthen the claim that the result above is not an artefact of smooth compounding.

It would still not establish probability. The period contains the recovery from the 1970s, globalisation, four decades of falling interest rates and the post 2009 bull market. The paper's conclusions rest on this paper's forward assumptions and on annex A, not on this annex.

Annex C: figures read from published sources rather than computed here

Stated so a reader knows which numbers in this paper rest on the author's reading of a published document rather than on a figure computed here, and where to go to check each. Most of the sources below are primary institutional publications; what they have in common is not their status but their route into this paper, which is quotation rather than calculation. **This is a limitation of the paper, not a list of pending work.**

- **OBR, Fiscal Risks and Sustainability, July 2026.** The 5% to 9% of GDP pension path and the earnings indexation variant. Read the report rather than coverage of it.
- **Dimson, Marsh and Staunton, Global Investment Returns Yearbook.** Cited for long run real returns. **A caution worth carrying:** the 2026 edition's headline 6.6% real equity figure appears to be the US series, and the world figure is the relevant one and materially lower. A reader comparing this paper's figures against a yearbook headline should confirm which series they are reading.
- **DMO gilt yields,** taken on the day of drafting, with the discrepancy between aggregators noted at section 3.5 unresolved.
- **Norges Bank Investment Management.** The mandate excluding domestic assets, and the expected real return underpinning the spending rule.
- **ONS.** Current public sector net financial liabilities as a share of GDP.

The historical series used for the testing at section 5A and Annex B is not in this list. It is computed from the source at Annex D and specified there in full.

Annex D: companion paper, data, attribution and reproducibility

The companion paper. *The Long Asset* sets out the constitutional architecture that rests on the findings here: how a fund of this kind should be owned, governed, reported and protected so that the arithmetic tested in this paper survives ordinary politics. It is conditional on the proposition tested here, and states so at its opening. The two papers form a self-contained pair; neither requires any other work of the author's.

No third-party data is reproduced in this paper. The tables report statistics computed by the author from the sources below. No dataset, series or extract is republished here or made available for download, and no proprietary or licensed commercial index data is used in any figure reported here.

Historical returns, exchange rates, inflation and interest rates

Computed from the Jordà-Schularick-Taylor Macrohistory Database, release 6, consulted 13 August 2026, available at macrohistory.net. The database is made available by its authors at no cost under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International licence. In accordance with the attribution terms, the database is cited as follows, the first for the dataset generally and the second for its rate-of-return series:

Òscar Jordà, Moritz Schularick and Alan M. Taylor (2017), "Macrofinancial History and the New Business Cycle Facts", in *NBER Macroeconomics Annual 2016*, volume 31, ed. Martin Eichenbaum and Jonathan A. Parker, Chicago: University of Chicago Press.

Òscar Jordà, Katharina Knoll, Dmitry Kuvshinov, Moritz Schularick and Alan M. Taylor (2019), "The Rate of Return on Everything, 1870-2015", *Quarterly Journal of Economics* 134(3), 1225-1298.

UK fiscal aggregates

Office for Budget Responsibility, historical public finances database and long-term economic determinants, used under the Open Government Licence v3.0. Contains public sector information licensed under the Open Government Licence v3.0.

Construction

National equity total returns in local currency are converted to sterling using contemporaneous exchange rates from the same source, equally weighted across the countries with data in each year, and deflated by UK consumer prices. The financing leg is the realised UK long-term government rate deflated by the same index. The index universe grows from eight countries in 1871 to sixteen by 2020; Canada and Ireland carry no equity data in the source and are excluded throughout. The exclusion is forced rather than selected: the source's return dataset covers sixteen advanced economies, as the citation above states, and any reader working from the published workbook necessarily arrives at the same sixteen. No result rests on a chosen start date: every figure uses the complete set of rolling windows available.

Declared analytical choices

Each is a choice rather than a measurement, and a reader disagreeing with any of them can locate what it moves:

- **equal weighting** rather than market-capitalisation weighting, which is unavailable without licensed data. Switching to GDP weighting moves the boundary by about 2.9 points;
- **unhedged sterling**, which embeds four decades of sterling depreciation. Annex B warns this should not be treated as a repeatable source of return, and hedged and unhedged constructions should both be reported;
- **consumer prices** rather than an alternative deflator;
- **the realised long-term government rate** as the financing proxy, in place of a market-observed issuance curve. This is the dominant unresolved choice, by a factor of roughly five over the weighting basis.

Reproducibility

These figures are not the output of a proprietary index or of any dataset requiring paid access. They are computed from a freely available academic database using the construction above, and any reader can obtain the same source and recompute them. No commercial index data is reproduced here.

This is a deliberate property rather than a limitation of budget. A result derived from a licensed index cannot be checked by a reader without that licence; a result derived from an open database and an openly described method can be checked by anyone. **The specification below is complete enough to reconstruct every historical figure in this paper and its companion**, and a reader wishing to test them should do so independently rather than take the author's implementation on trust. An independent reconstruction is a better check than a supplied one: it varies the implementation as well as re-running it, and where the two disagree the disagreement is itself informative.

Reconstruction specification. Contributions of one unit at each year end. Investment return applied to the opening balance. Drawdown taken as the stated share of realised real return before reinvestment, and zero where the return is negative. The financing leg accumulates the same unit contributions at the realised real financing rate. The reported net position is the fund less that accumulated financing cost. Windows are every complete run of the stated length available in the record, with no window excluded and no start date preferred. Boundaries are found by bisection on the distributable share, to the point at which the terminal net position changes sign.

Where a reader has access to a second long-run compilation, that is a materially stronger test than reproducing these figures from the same source, and section 7A states what it would have to show. Reproducing the arithmetic from one dataset confirms the arithmetic, not the dataset.

Licence

This paper is published for non-commercial purposes under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International licence. It may be shared and adapted, for non-commercial purposes, with attribution and under identical terms. That is the licence carried by the principal dataset on which the cited calculations depend, and it is adopted here by choice rather than as a statement that this paper is an adaptation of that dataset within the meaning of the licence. Whether computed statistics constitute an adaptation of a database or an analysis of it is a question this paper does not need to answer, and does not answer, because the same terms apply either way.